

Servlet – HTTP Status Codes

For each HTTP request and HTTP response, we have messages. The format of the HTTP request and HTTP response messages are similar and will have the following structure –

- An initial status line + CRLF
- CRLF = (Carriage Return + Line Feed i.e. New Line)
- Zero or more header lines + CRLF
- A blank line, i.e., a CRLF
- An optional message body like file, query data, or query output.

For example, a server response header looks like–

- HTML

```
HTTP/5.0 200 OK
```

```
Content-Type: text/html
```

```
Headers:
```

```
(Blank Line)
```

```
<!doctype HTML>
```

```
<html>
```

```
<head></head>
```

```
<body>
```

```
</body>
```

```
</html>
```

The status line consists of the HTTP version (HTTP/5 in the example), a status code (200 in the example), and a very short message corresponding to the status code (OK in the example). Here is a list of HTTP status codes and associated messages that might be returned from the Web Server –

Status Code	Message	Description
100	Continue	Only a part of the request has been received by the server, but as long as it has not been rejected, the client should continue with the request
101	Switching Protocols	The server switches protocol.
200	OK	The request is OK
201	Created	The request is complete, and a new resource is created
202	Accepted	The request is accepted for processing, but the processing is not complete.
203	Non-authoritative Information	
204	No content	
205	Reset content	
206	Partial content	
300	Multiple choices	A link list. The user can select a link and go to that location. Maximum five addresses

Status Code	Message	Description
301	Moved permanently	The requested page has moved to a new URL
302	Found	The requested page has moved temporarily to a new URL
303	See other	The requested page can be found under a different URL
304	Not Modified	
305	Use Proxy	
306	Unused	This code was used in a previous version. It is no longer used, but the code is reserved
307	Temporary redirect	The requested page has moved temporarily to a new URL
400	Bad request	The server did not understand the request
401	Unauthorized	The requested page needs a username and a password
402	Payment required	You cannot use this code yet
403	Forbidden	Access is forbidden to the requested page
404	Not found	The server cannot find the requested page.

Status Code	Message	Description
405	Method not found	The method specified in the request is not allowed.
406	Not Acceptable	The server can only generate a response that is not accepted by the client.
407	Proxy Authentication Required	You must authenticate with a proxy server before this request can be served.
408	Request Timeout	The request took longer than the server was prepared to wait.
409	Conflict	The request could not be completed because of a conflict.
410	Gone	The requested page is no longer available.
411	Length Required	The “Content-Length” is not defined. The server will not accept the request without it.
412	Precondition Failed	The precondition given in the request was evaluated as false by the server.
413	Request Entity Too Large	The server will not accept the request, because the request entity is too large.

Status Code	Message	Description
414	Request-url Too Long	The server will not accept the request, because the URL is too long. Occurs when you convert a “post” request to a “get” request with long query information.
415	Unsupported Media Type	The server will not accept the request, because the media type is not supported.
417	Expectation Failed	
500	Internal Server Error	The request was not completed. The server met an unexpected condition.
501	Not Implemented	The request was not completed. The server did not support the functionality required.
502	Bad Gateway	The request was not completed. The server received an invalid response from the upstream server.
503	Service Unavailable	The request was not completed. The server is temporarily overloading or down.
504	Gateway Timeout	The gateway has timed out.
505	HTTP Version Not Supported	The server does not support the “http protocol” version.

Methods to Set HTTP Status Code

The below methods can be used to set HTTP Status Code in your servlet program. These methods are available with the `HttpServletResponse` object.

S.No.	Method and Description
1	<code>public void setStatus (int statusCode)</code> This method sets an arbitrary status code. The <code>setStatus</code> method takes an <code>int</code> (the status code) as an argument. If your response includes a special status code and a document, be sure to call <code>setStatus</code> before actually returning any of the content with the <code>PrintWriter</code> .
2	<code>public void sendRedirect(String url)</code> This method generates a 302 response along with a <code>Location</code> header giving the URL of the new document
3	<code>public void sendError(int code, String message)</code> This method sends a status code (usually 404) along with a short message that is automatically formatted inside an HTML document and sent to the client.

HTTP Status Code Example

The code below is the example that would send a 407 error code to the client browser and the browser would show you the “Not Found!!!” message.

```
// Import required java libraries
import java.io.*;
import javax.servlet.*;
import javax.servlet.http.*;
import java.util.*;

// Extend HttpServlet class
public class showError extends HttpServlet {
    // Method to handle GET method request.
    public void doGet(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {

        // Set error code and reason.
        response.sendError(404, "Not Found!!!" );
    }

    // Method to handle POST method request.
    public void doPost(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {
        doGet(request, response);
    }
}
```

Now calling the above servlet would display the following result –

HTTP Status 404 – Not Found

Type Status Report

Message The requested resource [/Basic/servlet/showError] is not available

Description The origin server did not find a current representation for the target resource or is not willing to disclose that one exists.

Apache Tomcat/8.5.70